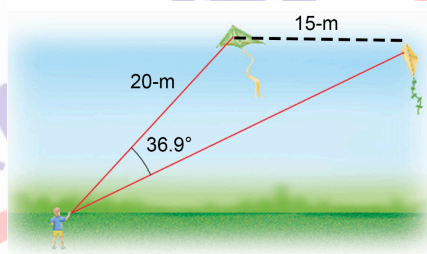


Cursillo Stewart

UPTP Entrance Simulation 2024

- Write your answers in the answer sheet.
- Some formulas are provided in the last page.

- Find $\sin^{-1}(\cos \frac{\pi}{3}) + \cos^{-1}(\sin \frac{5\pi}{4})$
 (A) $\frac{5\pi}{12}$ (B) $\frac{\pi}{6}$ (C) $\frac{11\pi}{12}$ (D) $\frac{19\pi}{12}$ (E) $-\frac{\pi}{12}$
- Please find the simple expression of $\frac{\sin(x) + \sin(5x)}{\cos(x) + \cos(5x)}$
 (A) $\cot(3x)$ (B) $\tan(x)$ (C) $\sec(x)$ (D) $\sin(3x)$ (E) *None of the above*
- If θ is in *Quadrant IV* and $\cos(\theta) = \frac{5}{7}$, and β is in *Quadrant III* and $\cot(\beta) = \sqrt{3}$. Find $\tan(\theta + \beta)$:
 (A) $\frac{15+6\sqrt{2}}{6\sqrt{6}+5\sqrt{3}}$ (B) $\frac{-6\sqrt{6}+5\sqrt{3}}{15+6\sqrt{2}}$ (C) $\frac{6\sqrt{6}+5\sqrt{3}}{15-6\sqrt{2}}$ (D) $\frac{15+6\sqrt{2}}{6\sqrt{6}+5\sqrt{3}}$ (E) *None of the above*
- A boy is flying two kites at the same time. He has a 20 m rope for one kite. The distance between the two kites is 15 m, and the angle between the ropes of the two kites is 36.9° . Find the length of the rope for the second kite.



- (A) 32.5 m (B) 22 m (C) 25 m (D) 27.5 m (E) *None of the above*
- Which of the following IS a solution for the trigonometric equation:
 $2\sin^2(\theta) = 2 + \cos(2\theta)$:
 (A) $\frac{5\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{7\pi}{3}$ (D) $\frac{3\pi}{2}$ (E) *None of the above*
 - In which quadrants $\sin(\theta)$. $\cos(\theta)$ have positive values?
 (A) *Quadrant II and Quadrant IV*
 (B) *Quadrant I and Quadrant II*
 (C) *Quadrant I and Quadrant III*
 (D) *Quadrant IV*
 (E) *None of the above*
 - Find the amplitude, period, phase and horizontal shift for curve $y = 50 \sin(8t + \frac{\pi}{2})$
 (A) Amplitude: 50; Period: 8π ; H.S.: $\frac{\pi}{2}$; Phase: $\frac{\pi}{16}$
 (B) Amplitude: 50; Period: $\frac{\pi}{4}$; H.S.: $\frac{\pi}{2}$; Phase: $\frac{\pi}{2}$
 (C) Amplitude: 50; Period: $\frac{\pi}{4}$; H.S.: $\frac{\pi}{8}$; Phase: $\frac{\pi}{8}$
 (D) Amplitude: 50; Period: $\frac{\pi}{4}$; H.S.: $\frac{\pi}{16}$; Phase: $\frac{\pi}{2}$

(E) None of the above

8. If $135^\circ < \theta < 140^\circ$, let $a = 1 + \sin(\theta)$; $b = \frac{\tan(\theta)+1}{\tan(\theta)}$; $c = \log(\theta)$; $d = e^\theta$. To find the order of a, b and c , which of the following is correct?

(A) $b > a > c > d$

(B) $d > c > a > b$

(C) $a > b > d > c$

(D) $d > a > c > b$

(E) None of the above

9. Find $\sin(75^\circ) + 2 \cos(75^\circ)$

(A) $\frac{\sqrt{6}+\sqrt{2}}{2}$

(B) $\frac{3\sqrt{6}-\sqrt{2}}{4}$

(C) $\frac{\sqrt{6}-\sqrt{2}}{2}$

(D) $\frac{\sqrt{6}+3\sqrt{2}}{2}$

(E) 0

10. For a circle with radius 10, find the terminal point determined by $\frac{11\pi}{6}$.

(A) $(-5\sqrt{3}, -5)$

(B) $(5\sqrt{3}, 5)$

(C) $(-5\sqrt{3}, 5)$

(D) $(5\sqrt{3}, -5)$

(E) None of the above

11. The value of $\log(0.00336)$ can be round to two decimal places. What is its approximate value? (Hint: $\log(2) = 0.301$, $\log(3) = 0.477$, $\log(7) = 0.845$)

(A) -1.38

(B) -2.47

(C) -3.36

(D) -0.54

(E) None of the above

12. The vector $X \langle a, b \rangle$ and $Y \langle c, d \rangle$ are two real vectors. Which of the following conditions must be true for X and Y to be perpendicular?

(A) $ad + bc = 1$

(B) $ad - bc = 0$

(C) $ac + bd = 1$

(D) $ac - bd = 0$

(E) $ac + bd = 0$

13. Convert the following polar equation $r = 3\sin^2(2\theta)$ to rectangular coordinates.

(A) $(x^2 + y^2)^5 = 144x^4y^4$

(B) $(x^2 + y^2) = 12x^2y^2$

(C) $(x^2 + y^2)^5 = 144x^5y^5$

(D) $(x^2 - y^2)^3 = 12x^3y^3$

(E) None of the above

14. Which of the following statements is FALSE regarding the laws of logarithm?

(A) $\log(AB) = \log A + \log B$

(B) $\log \frac{A}{B} = \log A - \log B$

(C) $\log A^c = c \log A$

(D) $\log_{1/a^2} B = \log_a B^{1/2}$

(E) $\log_A B = \frac{\log_c B}{\log_c A}$

15. A plane intersects the coordinates axes at the points $(2, 3, 0)$, $(0, 4, 2)$ and $(-1, 2, 1)$.

The obtained plane equation is $x + ay + bz = c$. What is the value of b ?

(A) $-\frac{4}{3}$

(B) $\frac{5}{3}$

(C) -2

(D) 1

(E) None of the above

16. Solve the following exponential equation $e^x + 15e^{-x} - 8 = 0$

(A) $x_1 = \ln(2), x_2 = \ln(5)$

(B) $x_1 = \ln(2), x_2 = \ln(3)$

(C) $x_1 = \ln(3), x_2 = \ln(5)$

(D) $x_1 = \ln(3)$

(E) No real solutions

17. The noise level at a jazz concert was measured at 120.15 dB. The noise level at a street of NY was measured at 91.7 dB. Find the ratio of the intensity of the jazz concert to that of the street's noise. (Hint: $\log(7) = 0.845$)

- (A) $\frac{I_1}{I_2} = 845$ (B) $\frac{I_1}{I_2} = 84.5$ (C) $\frac{I_1}{I_2} = 70$ (D) $\frac{I_1}{I_2} = 700$ (E) $\frac{I_1}{I_2} = 560$

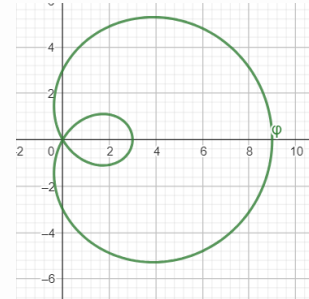
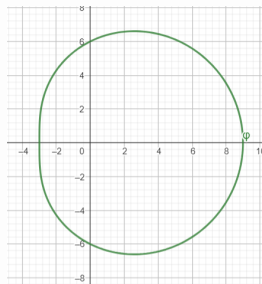
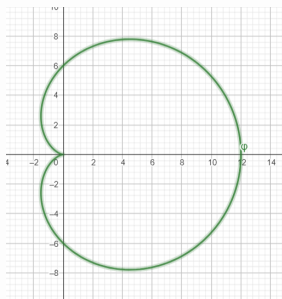
18. Which of the following options best represents the graph of the polar equation

$$r = 6 + 3 \cos \theta.$$

(A)

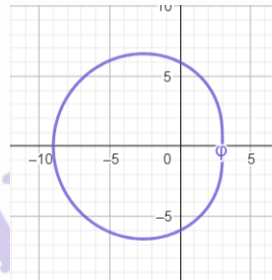
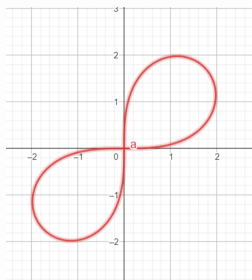
(B)

(C)



(D)

(E)



19. Determine the area of the triangle formed by the points $(-3, \frac{\pi}{6})$, $(4, \frac{3\pi}{4})$ and the pole in the polar coordinates.

- (A) $\frac{3\sqrt{2}}{2}(\sqrt{3} + 1)$ (B) $\frac{\sqrt{2}}{2}(\sqrt{3} - 1)$ (C) $\frac{3\sqrt{2}}{2}(\sqrt{3} + 1)$ (D) $\frac{3}{2}(\sqrt{3} + 1)$
(E) None of the above

20. A vector makes an angle of 60° with the positive x -axis and an angle of 135° with the positive z -axis. Determine the value of the obtuse angle β that the vector makes with the positive y -axis.

- (A) $\beta = 150^\circ$ (B) $\beta = 60^\circ$ (C) $\beta = 120^\circ$ (D) $\beta = 105^\circ$ (E) $\beta = 45^\circ$

21. A rational function is defined as:

$$r(x) = \frac{x^3 + 4}{2x^2 + x - 1}$$

What is the slant asymptote of $r(x)$?

- (A) $\frac{x}{2} - \frac{1}{4}$ (B) $\frac{x}{2} - 1$ (C) $x - \frac{1}{4}$ (D) $\frac{x}{4} - \frac{1}{2}$ (E) None of the above

22. Which of the following functions is an even function?

(A) $f(x) = \frac{x+1}{3+x^2}$

(B) $f(x) = 2x - x^2$

(C) $f(x) = \frac{3-5x^2}{7+x^6}$

(D) $f(x) = 6x^5 + 5x$

(E) None of the above

23. Find a polynomial with degree 4, zeros i and $1 + i$, and constant term 12.

- (A) $2x^4 - 6x^3 + 6x^2 - 4x + 12$ (B) $x^4 - 2x^3 - 3x^2 - 2x + 12$
(C) $6x^4 + 12x^3 - 24x^2 + 12x + 12$ (D) $6x^4 - 12x^3 + 18x^2 - 12x + 12$
(E) None of the above

24. A function is given $f(x) = \frac{6x-3}{2x+7}$. What is the inverse function of $f(x)$?

- (A) $f^{-1}(x) = \frac{3-7x}{2x+6}$ (B) $f^{-1}(x) = -\frac{3+7x}{2x-6}$ (C) $f^{-1}(x) = \frac{-3+7x}{6-2x}$ (D) $f^{-1}(x) = \frac{-3-7x}{2x-4}$
(E) None of the above

25. Mary inherits \$100,000 and invests it in two certificates of deposit. One certificate pays 6% and the other pays $4\frac{1}{2}\%$ simple interest annually. If Mary's total interest is \$5025 per year, how much money is invested at the $4\frac{1}{2}\%$ rate?

- (A) \$35,000 (B) \$45,000 (C) \$65,000 (D) \$85,000 (E) \$95,000

26. Which one is the solution to the following inequality?

$$\frac{1+x}{1-x} \geq 1$$

- (A) $[0, 1]$ (B) $(-\infty, 0) \cup (0, 1)$ (C) $(0, 1)$ (D) $(-\infty, 0) \cup (1, \infty)$
(E) None of the above

27. What amount of a 40% acid solution must be mixed with 150 mL of a 10% solution to produce a 20% solution?

- (A) 95 mL (B) 85 mL (C) 75 mL (D) 105 mL (E) 115 mL

28. A function is given

$y = -7(x - 3)^2 - 4$. What are the indicated transformation applied to the graph of $f(x) = x^2$ in the given order? I

- (A) Shrunk vertically by a factor of 7, shifted upward 4 units, shifted 3 units to the left and reflected about the y -axis.
(B) Stretched vertically by a factor of 7, shifted downward 4 units, shifted 3 units to the left and reflected about the x -axis.
(C) Shrunk vertically by a factor of 7, shifted upward 4 units, shifted 3 units to the right and shifted reflected about the y -axis.
(D) Stretched vertically by a factor of 7, shifted downward 4 units, shifted 3 units to the right, and reflected about the x -axis.
(E) None of the above

29. Find a line that is perpendicular to the line $7x - 5y = 8$ and passed through the point at $(-1, -3)$.

- (A) $7y + 5x = -26$ (B) $5y - 7x = 26$ (C) $7y - 5x = -26$ (D) $7y + 5x = 26$
(E) None of the above

30. Find the area of the region that lies outside the circle $x^2 - 8x + y^2 - 4y - 12 = 0$ but inside the circle $x^2 + y^2 - 10x - 6y - 30 = 0$.

- (A) 32 (B) 64π (C) 32π (D) 64 (E) 36π

31. Matrices A and B are defined as $A = \begin{bmatrix} 1 & 2 & -3 \\ 7 & -5 & 5 \\ 5 & 2 & -10 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 5 & 3 \\ -2 & 8 & 4 \\ -1 & 1 & -2 \end{bmatrix}$. Solve BA .

(A) $\begin{bmatrix} 0 & 18 & 17 \\ 12 & 0 & -9 \\ 11 & 31 & 43 \end{bmatrix}$

(B) $\begin{bmatrix} 0 & 18 & -7 \\ 24 & 0 & -9 \\ 13 & 11 & 43 \end{bmatrix}$

(C) $\begin{bmatrix} 51 & -17 & -8 \\ 12 & 0 & -9 \\ -4 & 13 & 28 \end{bmatrix}$

(D) $\begin{bmatrix} 51 & -17 & -8 \\ 74 & -36 & 6 \\ -4 & -11 & 28 \end{bmatrix}$

(E) None of the above

32. Find the area of the region in the coordinate plane that satisfies both the inequalities

$$x^2 + y^2 \leq 36 \text{ and } y \geq |x - 6|$$

- (A) $18(1 - \pi)$ (B) $18(1 + \pi)$ (C) $18 - 9\pi$ (D) $9(\pi - 2)$ (E) $9(\sqrt{3} - \pi)$

33. The matrix A is defined as $A = \begin{bmatrix} 7 & -5 & 1 \\ 8 & 4 & 3 \\ -12 & 25 & -1 \end{bmatrix}$. Evaluate the determinant $|A|$.

(A) -156

(B) 145

(C) -165

(D) 96

(E) -96

34. For the partial fraction decomposition of the expression

$$\frac{x^2+6}{(x-2)(x+3)^2} = \frac{A}{(x-2)} + \frac{B}{(x+3)} + \frac{C}{(x+3)^2}.$$
 Find the value of B .

(A) $-\frac{2}{5}$

(B) $\frac{2}{5}$

(C) $\frac{3}{5}$

(D) -3

(E) -2

35. A parabola has a maximum point at $(x, y) = (-1, 2)$ and passes through the point $A(1, 1)$. Find the parabola equation.

(A) $(x + 1)^2 = 4(y - 2)$

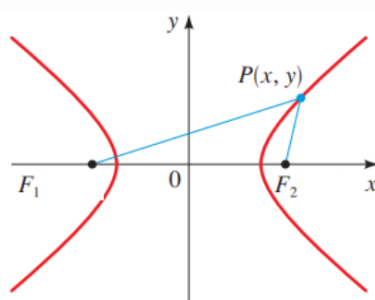
(B) $(x + 1)^2 = -2(y - 2)$

(C) $(x + 1)^2 = 4(2 - y)$

(D) $(y + 1)^2 = 4(x - 2)$

(E) $(x - 1)^2 = 2(y + 2)$

36. Below is the figure of a hyperbola in which F_1 and F_2 are its foci and P is a point on the hyperbola. Which one below is hyperbola's unique property where c is a constant where $\overline{PF_1}$ denotes the distance between point P and F_1 ?



(A) $|\overline{PF_1} + \overline{PF_2}| = c$

(B) $|\overline{PF_1} - \overline{PF_2}| = c$

(C) $|\overline{PF_1} \times \overline{PF_2}| = c$

(D) $|\overline{PF_1}| = c$

(E) None of the above

37. A hyperbola has the equation $4(x - 3)^2 - 16y^2 = 64$. Which of the following answers is correct?

(A) Center at $(-3, 0)$

(B) x - intercept at $x = 1$

(C) One asymptote at $y = \frac{3}{2} - \frac{x}{2}$

(D) Length of the transverse axis is 4

(E) None of the above

38. The second and fourth terms of an arithmetic sequence is given, $a_2 = 68$ and $a_4 = 4$.
Find the value of the sixth term.
(A) -32 (B) -60 (C) 0 (D) -24 (E) -64
39. The purchase value of an office computer is \$10,500. Its annual linear depreciation is \$723. Find the value of the computer after 6 years.
(A) \$7562 (B) \$8435 (C) \$6162 (D) \$6352 (E) *None of the above*
40. Find the partial sum of the sequence $5120 + 2560 + 1280 + \dots + 20$
(A) **10220** (B) 10200 (C) 10080 (D) 10240 (E) 10260

Note: Some formulae for your references, derive the others when needed

Base of natural logarithm: $e=2.718$

Continuously compound interest: $A=Pe^{rt}$, A : yield, P : capital, r : yearly interest rate, t : number of years

Addition/subtraction formulae: $\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$

$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$$

Half angle formulae: $\sin\left(\frac{\alpha}{2}\right) = \pm\sqrt{\frac{1-\cos\alpha}{2}}$, $\cos\left(\frac{\alpha}{2}\right) = \pm\sqrt{\frac{1+\cos\alpha}{2}}$

Product-to-sum: $\sin\alpha\cos\beta = \frac{1}{2}[\sin(\alpha + \beta) + \sin(\alpha - \beta)]$

$$\cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha + \beta) + \cos(\alpha - \beta)]$$

De Moivre's Theorem: if $z = r(\cos\theta + i\sin\theta)$, then for any integer n , $z^n = r^n(\cos(n\theta) + i\sin(n\theta))$

2×2 Matrix inverse formula: $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Standard equation of ellipse: $\frac{(x-x_0)^2}{a^2} + \frac{(y-y_0)^2}{b^2} = 1$ where (x_0, y_0) is the center of the ellipse

Sum of arithmetic series: $S_N = \sum_{n=1}^N a_n = \frac{(a_1+a_N)N}{2}$, with $a_n = a_1 + (n-1)d$, d : common difference

Sum of geometric series: $S_N = \sum_{n=1}^N a_n = \frac{a_1(r^N-1)}{(r-1)}$, with $a_n = a_1r^{n-1}$, r : common ratio

θ	0°	30°	36.9°	45°	60°	90°
$\sin\theta$	0	$\frac{1}{2}$	0.6	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	0.8	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0